

AMBER GARDENS



Visual: project identity graphic generated from the supplied project list. Official project images are available through the source links below.

Why is this project a best practice?

Challenge

The project addresses the difficulty of scaling low-energy and healthy-building principles beyond a single dwelling. It had to combine density or phased delivery with reliable energy performance, good indoor conditions, landscape quality, mobility and construction cost, while keeping the sustainability measures practical for many future occupants.

Solution

The response combines bioclimatic design and high-performance envelopes developed toward Passive House-level efficiency; photovoltaics, heat pumps, radiant heating/cooling and heat-recovery ventilation; and non-toxic and sustainably sourced materials, native planting, common gardens, playgrounds and electric-vehicle charging. These measures were brought together through an integrated design approach and, where applicable, the Green Homes, nZEB or Passive House performance framework.

Replicability

The approach can be scaled through repeatable specifications for envelope, MEP systems, materials, commissioning, mobility and landscape. Replication is strongest when the developer applies the same measurable requirements to every phase and verifies them at handover, rather than treating certification as a one-off design exercise.

Key facts and figures

Location	Strada 1 Decembrie 132, Tunari, Ilfov County, Romania
Building use / function	Low-density residential community of detached villas
Project type (new build, refurbishment, extension)	New build; phased development
Plot area / Floor area	Certified phase reported at approximately 6,240 m ² ; 21 completed villas and about 60 planned in the toolkit
Year / phase	First certified phase completed in 2014; later phases continued
Climate zone	Temperate continental
Thematic focus / innovation	Passive-house bioclimatic design; renewable energy; heat pumps; radiant systems; healthy materials and shared green space
Investor / building owner	Alesonor
Architect	Not publicly disclosed in the consulted sources
Other involved stakeholders	Romania Green Building Council; Passive House and building-services specialists; landscape team
Contact person	Alesonor - https://www.ambergardens.ro/
Further information	https://green-homes.org/property/amber-gardens-tunari/ https://www.rogbc.org/proiecte-certificate-ro gbc?lang=en https://www.rogbc.org/_files/ugd/40ee3e_b5d91d7a3b71422cb98cd7a289b89c79.pdf https://www.ambergardens.ro/

Amber Gardens is a new build; phased development project in Strada 1 Decembrie 132, Tunari, Ilfov County, Romania. It functions as low-density residential community of detached villas. Its main best-practice contribution is One of Romania's earliest certified green detached-home communities and an early near-net-zero market example. The documented strategy brings together bioclimatic design and high-performance envelopes developed toward Passive House-level efficiency, photovoltaics, heat pumps, radiant heating/cooling and heat-recovery ventilation, and non-toxic and sustainably sourced materials, native planting, common gardens, playgrounds and electric-vehicle charging. Unit counts vary by phase and source; the toolkit reports 21 completed villas and 60 planned, while later project pages cover different certified phases. Public-source research was reviewed in July 2026; data that vary by phase are explicitly flagged in this sheet.

What was done differently?

1. Performance concept - Bioclimatic design and high-performance envelopes developed toward passive house-level efficiency.
2. Integrated systems and operation - Photovoltaics, heat pumps, radiant heating/cooling and heat-recovery ventilation.
3. Wider value - Non-toxic and sustainably sourced materials, native planting, common gardens, playgrounds and electric-vehicle charging.

What was achieved?

Impact

Environmental impact: The project reduces energy and associated emissions primarily by lowering demand and then using efficient or renewable systems. Its material, landscape, waste or mobility measures add benefits beyond operational energy where documented. A verified whole-life carbon assessment was not publicly available, so the impact is described qualitatively unless a source provides a specific figure.

Economic impact: Lower energy demand can reduce utility exposure and improve long-term operating-cost predictability. Standardised high-performance specifications and third-party certification can also support quality assurance, market value and access to green-finance products. Public sources do not provide a complete investment, payback or life-cycle-cost analysis for this project.

Social impact: The main social benefits are stable indoor temperatures, better air quality, daylight and acoustic comfort, together with safer and more attractive shared or outdoor spaces where these form part of the project. The project also raises market awareness by making high-performance housing visible to residents, designers, contractors and investors.

Key Learnings

Success factors: Key success factors are an integrated design process, a clear performance framework, early coordination of envelope and building services, selection of competent suppliers, quality control during construction and commissioning. Transparent documentation and communication with occupants are essential to preserve the intended performance in use.

Lessons for replication: Replication should start with an energy and comfort target, not a catalogue of technologies. Design teams should verify thermal bridges, airtightness, ventilation, controls and renewable sizing together; record assumptions by project phase; commission systems; and monitor early operation. Where public figures differ by phase, the final as-built dataset must become the single source of truth.